Last time we estimated the displacement from a given velocity function, and found that the displacement should be given by the "area under the graph" of the velocity function. We now begin to make this notion precise.

1 Area

Given a function $f : [a,b] \to \mathbb{R}$, we wish to associate a number $\int_a^b f$, called the *integral of f from a to b*. When f is non-negative, $\int_a^b f$ represents the "area" of the region in the plane bounded above by the graph of $y = f(x)$, bounded below by the x-axis, bounded on the left by $x = a$ and bounded on the right by $x = b$. We denote this region by $R(f,a,b)$.

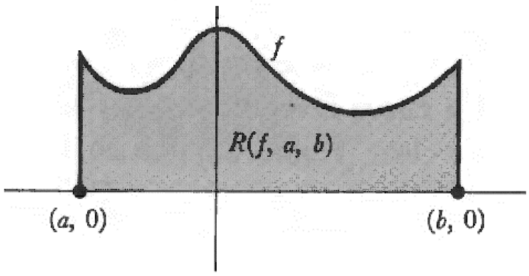


Figure 1: When $f(x) \geq 0$ on $[a,b]$, $\int_a^b f =$ the area of $R(f,a,b)$.

If f is not non-negative, such as the function graphed below, then $\int_a^b f$ will represent the difference of the area of the area above the x-axis and the area below the x-axis (the "algebraic area" of $R(f,a,b)$).

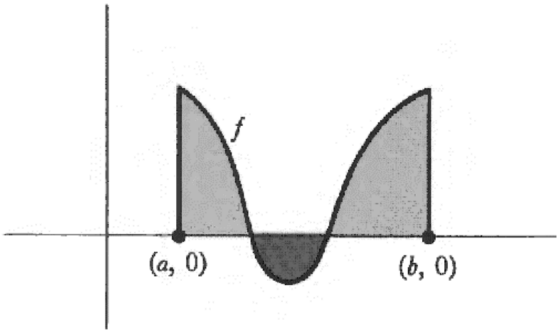


Figure 2: $\int_a^b f =$ (area above the x-axis) - (area below the x-axis).

In elementary geometry, area is generally defined as the number of unit squares which fit in a region, and formulas are derived for the areas of many plane figures. However, this definition is inadequate for all but the simplest regions. For instance, a circle of radius 1 supposed has an area of π, but it is unclear what "π squares" means. Even if we consider a circle of radius $\frac{1}{\sqrt{\pi}}$, which supposedly has area 1, it is hard to say in what way a unit square fits in this circle, since it does not seem possible to divide the unit square into pieces which can be arranged to form a circle.

2 Properties of the Integral

Before setting out to define $\int_a^b f$, we first ask: what are some reasonable properties that the "area" of $R(f,a,b)$ should satisfy?

1. Consider a constant function $f(x) = c > 0$. Sketch the region $R(f,a,b)$. What is its area?

2. Suppose $0 \leq f_1(x) \leq f_2(x)$ for all $x \in [a,b]$. Sketch two possible regions $R(f_1,a,b)$ and $R(f_2,a,b)$ on the same axes. How should the areas compare?

3. Let c be any point such that $a \leq c \leq b$ and let $f(x) \geq 0$ for all $x \in [a,b]$. Draw a possible region $R(f,a,b)$ and indicate the regions $R(f,a,c)$ and $R(f,c,b)$. How should the area of $R(f,a,b)$ be related to the areas of $R(f,a,c)$ and $R(f,c,b)$?

♢ **Check your answers with your instructor before moving on.**

We now define $\int_a^b f$ axiomatically, using the properties observed above.

Definition 2.1 (The integral)**.** The *integral* is a mapping $f \mapsto \int_a^b f$ which assigns a function $f : [a,b] \to \mathbb{R}$ to a real number $\int_a^b f$ in such a way that satisfies the following three axioms:

(I1) If $f = c$ is a constant function, then $\int_a^b c = c(b-a)$.
(I2) If $f_1(x) \leq f_2(x)$ for all $x \in [a,b]$, then $\int_a^b f_1 \leq \int_a^b f_2$.
(I3) If $a \leq c \leq b$, then $\int_a^b f = \int_a^c f + \int_c^b f$.

These three axioms already imply many other properties we want the integral to have. For example,

4. Show (I3) implies that for any $f : [a,b] \to \mathbb{R}$ and any $c \in [a,b]$, $\int_c^c f = 0$.

Before exploring other properties, note that we have not yet explicitly addressed the domain of the mapping $f \mapsto \int_a^b f$. That is, for which functions $f : [a,b] \to \mathbb{R}$ do we expect $\int_a^b f$ to be defined? A little thought shows that it is not plausible for the integral to be defined for *all* functions.

Example 2.2. If f is the function $f : [0,1] \to \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ for $x \neq 0$ (and $f(0) = c$ for some $c \in \mathbb{R}$), then $R(f,a,b)$ would has *infinite* area, so $\int_a^b f$ is not defined.

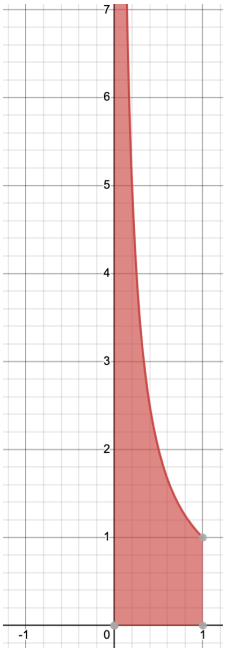


Figure 3: A region $R(f,a,b)$ with infinite area.

The previous example shows that for $\int_a^b f$ to be defined, then $f : [a,b] \to \mathbb{R}$ should at least be *bounded* (recall that $f : [a,b] \to \mathbb{R}$ is *bounded* if there exists $M \geq 0$ such that $|f(x)| \leq M$ for all $x \in [a,b]$). [1] However, we still don't expect that $\int_a^b f$ should exist for *all* bounded functions, as there exist bounded functions that are so pathological that the "area of the region $R(f,a,b)$" doesn't have any meaning.

Example 2.3. The function $f : [a,b] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational,} \\ 0, & \text{if } x \text{ is irrational,} \end{cases}$$

is bounded, but is discontinuous at every point. Its graph is therefore so crazy that the "area of the region $R(f,a,b)$" doesn't make any intuitive sense, so we don't expect $\int_a^b f$ to exist for such a function.

In particular, if f is continuous then we expect that $\int_a^b f$ should exist. However, while continuity should be sufficient for existence of $\int_a^b f$, it is not necessary.

5. Consider the function $f : [0,2] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & \text{if } 0 \leq x < 1, \\ 2, & \text{if } 1 \leq x \leq 2. \end{cases}$$

   (a) Show that f is discontinuous on $[0,2]$.

   (b) Explain why $\int_0^2 f$ should exist and find its value.

Definition 2.4 (Integrable function)**.** Denote the domain of the mapping $f \mapsto \int_a^b f$ by $\mathscr{R}[a,b]$. Elements of $\mathscr{R}[a,b]$ are called *integrable functions*.

By the discussion above, we will assume (that is, we will also take to be an axiom) that the set $\mathscr{R}[a,b]$ of integrable functions on $[a,b]$ properly contains the set $\mathscr{C}[a,b]$ of continuous functions on $[a,b]$ and is properly contained in the set $\mathscr{B}[a,b]$ of bounded functions on $[a,b]$; i.e., we assume that

$$\mathscr{C}[a,b] \subset \mathscr{R}[a,b] \subset \mathscr{B}[a,b].$$

[1] E.g., $\sin(x)$ is bounded since $|\sin(x)| \leq 1$ for all $x \in \mathbb{R}$.

3 The Fundamental Theorem of Calculus

It turns out that, for *continuous* functions, there is a *unique* choice of mapping $f \mapsto \int_a^b f$ which satisfies axioms (I1) - (I3), assuming such a mapping exists.

Before proving this assertion, it will be convenient to make the following definition. So far, $\int_a^b f$ is only defined for $a \leq b$. We now define $\int_a^b f = -\int_b^a f$ if $a > b$. With this definition, $\int_a^b f = \int_a^c f + \int_c^b$ for *all* a, b, and c, even if $a < c < b$ is not true (the proof is a rather tedious case-by-case check, which we omit).

Theorem 3.1 (Fundamental Theorem of Calculus)**.** *Let $f \in \mathscr{R}[a,b]$ be any integrable function. For $x \in [a,b]$ define $\mathscr{F} : [a,b] \to \mathbb{R}$ by $\mathscr{F}(x) = \int_a^x f$. Then:*

(a) The function $\mathscr{F}$ is continuous at every $c \in [a,b]$.

(b) If f is continuous at $c \in [a,b]$, then $\mathscr{F}$ is differentiable at c, and $\mathscr{F}'(c) = f(c)$.

(c) If f is continuous on $[a,b]$ and F is any antiderivative of f (i.e., a function $F : [a,b] \to \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in [a,b]$), then $\int_a^b f = F(b) - F(a)$.

If $f : [a,b] \to \mathbb{R}$ is continuous, then part (b) of the FTC shows that $\int_a^b f = \mathscr{F}(b)$, where $\mathscr{F}$ is an antiderivative for f with $\mathscr{F}(a) = \int_a^a f = 0$. If F is any other antiderivative for f with $F(a) = 0$, then $F(x) = \mathscr{F}(x) + c$, where c is a constant. By part (c) of the FTC, $\mathscr{F}(x) = \int_a^x f = F(x) - F(a) = F(x) - 0 = F(x)$ for each $x \in [a,b]$, and therefore $F = \mathscr{F}$, proving that there is a unique choice for $\int_a^b f$ if f is continuous (again, assuming that such a mapping $f \mapsto \int_a^b f$ exists). We will prove the FTC next time, but for now we take a look at a couple of examples.

Example 3.2. Consider the function $f(x) = x^n$, where $n \in \mathbb{N}$. It's easy to see that every antiderivative of f is of the form $F(x) = \frac{x^{n+1}}{n+1} + c$ for some constant $c \in \mathbb{R}$. Since f is continuous, by part (c) of the FTC we have

$$\int_a^b f = \frac{b^{n+1}}{n+1} - \frac{a^{n+1}}{n+1}.$$

6. Use the FTC to find $\int_a^b f$ for

   (a) $f(x) = \cos(x)$,

   (b) $f(x) = \sin(x)$.